

තොරතුරු තාක්ෂණ නිලධාරී විභාගය – Advanced Deep Notes

Sinhala + English Together (Nothing Removed)

1. ICT (Information and Communication Technology) Deep Notes

තොරතුරු හා සන්නිවේදන තාක්ෂණය

ICT යනු කුමක්ද? (What is ICT?)

තොරතුරු (Information) එකතු කිරීම (Collecting), ගබඩා කිරීම (Storing), සැකසීම (Processing), ආරක්ෂා කිරීම (Protecting), ප්‍රතිසාධනය කිරීම (Retrieving) සහ හුවමාරු කිරීම (Exchanging) සඳහා භාවිත කරන තාක්ෂණය ICT ලෙස හැඳින්වේ.

ICT Components Detailed

ICT කොටස් විස්තරාත්මකව

1. Hardware (දෘඩාංග)

පරිගණකයේ භෞතික කොටස් (Physical Parts of a Computer).

Hardware Categories (දෘඩාංග වර්ග)

Input Devices (ආදාන උපකරණ)

පරිගණකයට Data/Input ලබාදීමට භාවිත කරයි.

Examples (උදාහරණ)

Device	Function
Keyboard	Data Input
Mouse	Cursor Control
Scanner	Scan Documents
Webcam	Capture Video
Microphone	Audio Input

Output Devices (ප්‍රතිදාන උපකරණ)

පරිගණකයෙන් තොරතුරු ලබාදෙයි.

Device	Function
Monitor	Display Output
Printer	Print Documents
Speaker	Audio Output
Projector	Large Display

Storage Devices (ගබඩා උපකරණ)

දත්ත ගබඩා කිරීමට භාවිත කරයි.

Device	Function
HDD	Magnetic Storage
SSD	Fast Solid Storage
Pen Drive	Portable Storage
CD/DVD	Optical Storage

Processing Devices (සැකසුම් උපකරණ)

Data Process කිරීම සිදු කරයි.

Example

- CPU (Central Processing Unit)

CPU (Central Processing Unit)

මධ්‍යම සැකසුම් ඒකකය

Computer එකේ “Brain of the Computer” ලෙස හැඳින්වේ.

Main Parts of CPU (CPU කොටස්)

ALU (Arithmetic Logic Unit)

ගණිත හා තාකරීක ඒකකය

Functions

- Arithmetic Calculations (ගණිතමය ගණනය)
- Logical Operations (තාකරීක ක්‍රියා)

Examples

- Addition
- Subtraction
- Comparisons

CU (Control Unit)

පාලන ඒකකය

සියලු පරිගණක ක්‍රියා පාලනය කරයි (Controls all computer operations).

Functions

- Instructions control කිරීම
 - Data flow control කිරීම
 - Components coordinate කිරීම
-

Registers (රෙජිස්ටර්)

Temporary High-Speed Storage.

CPU එක තුළ ඇති වේගවත් මතක කොටස්.

Memory Types (මතක වගර්)

Primary Memory (ප්‍රාථමික මතකය)

CPU එකට directly access කළ හැක.

Types

- RAM
 - ROM
-

RAM (Random Access Memory)

Features

- Temporary Memory (තාවකාලික මතකය)
- Volatile Memory
- Fast Access
- Programs temporarily store කරයි

Advantages

- Fast processing
- Easy access

Disadvantages

- Power off වුවහොත් data නැතිවේ
-

ROM (Read Only Memory)

Features

- Permanent Memory
- Non-Volatile
- Boot Instructions store කරයි

Advantages

- Data permanently stored
- Stable

Disadvantages

- Hard to modify
-

RAM vs ROM

RAM	ROM
Temporary	Permanent
Volatile	Non-Volatile
Fast	Slower
Editable	Hard to Edit

Secondary Storage (ද්විතීයික ගබඩා)

Long-term data storage සඳහා භාවිත කරයි.

Examples

- HDD
 - SSD
 - Pen Drive
 - Memory Card
 - CD/DVD
-

2. Software Deep Notes

මෘදුකාංග

Software යනු කුමක්ද?

Computer එකට instructions ලබාදෙන programs.

Types of Software

මෘදුකාංග වගර්

1. System Software (පද්ධති මෘදුකාංග)

Hardware control කරයි.

Examples

- Operating Systems
- Drivers

- Utility Programs
-

2. Utility Software (උපකාරක මෘදුකාංග)

System maintenance සඳහා භාවිත කරයි.

Examples

- Antivirus
 - Backup Tools
 - Disk Cleanup
 - Disk Defragmenter
-

3. Application Software (යෙදුම් මෘදුකාංග)

User Tasks සඳහා භාවිත කරයි.

General Purpose Software

සාමාන්‍ය භාවිතය සඳහා.

Examples

- MS Word
 - Excel
 - Browsers
-

Customized Software

Specific organization එකකට develop කරන software.

Examples

- School Management System
- Hospital Management System
- Payroll System

3. Operating System Deep Theory

මෙහෙයුම් පද්ධති

Operating System යනු කුමක්ද?

Hardware සහ Software manage කරන System Software එක.

Functions of Operating System

මෙහෙයුම් පද්ධතියේ කාර්යයන්

Process Management

Running Programs manage කරයි.

Functions

- CPU allocation
 - Multitasking support
-

Memory Management

Memory allocate කරයි.

Functions

- RAM management
- Memory allocation/deallocation

File Management

Files සහ folders control කරයි.

Functions

- Create files
- Delete files
- Organize folders

Device Management

Hardware devices manage කරයි.

Examples

- Printer
- Keyboard
- Mouse

Security Management

Unauthorized access prevent කරයි.

Methods

- Passwords
- User Accounts
- Permissions

Types of Operating Systems

Single User Operating System

එක් User කෙනෙකුට පමණක් support කරයි.

Example

- MS-DOS
-

Multi User Operating System

Multiple Users support කරයි.

Example

- UNIX
 - Linux
-

Real Time Operating System (RTOS)

Instant response systems.

Used In

- Medical Equipment
 - Air Traffic Control
 - Industrial Systems
-

4. SDLC Deep Notes

පද්ධති සංවර්ධන ජීවන චක්‍රය

SDLC Importance

- Errors reduce කරයි

- Quality improve කරයි
 - Documentation improve කරයි
 - Easier maintenance
-

SDLC Steps

1. Planning
 2. Analysis
 3. Design
 4. Coding
 5. Testing
 6. Implementation
 7. Maintenance
-

1. Planning (සැලසුම් කිරීම)

Project objectives determine කිරීම.

Feasibility Study

Technical Feasibility

Technology available ද?

Economic Feasibility

Cost acceptable ද?

Operational Feasibility

Users use කරන්න පුළුවන්ද?

Legal Feasibility

Legal requirements meet කරනවද?

2. Analysis (විශ්ලේෂණය)

Current system problems identify කිරීම.

Methods Used

- Interviews
 - Questionnaires
 - Observation
 - Document Analysis
-

3. Design (සැලසුම් කිරීම)

New system structure prepare කිරීම.

Design Types

Input Design

Data input methods.

Output Design

Reports and outputs.

Database Design

Database structure.

Interface Design

User interface.

Good Design Features

- User Friendly
 - Secure
 - Reliable
 - Efficient
-

4. Coding (කේතනය)

Programming Languages භාවිතයෙන් system develop කිරීම.

Languages

- Java
 - Python
 - PHP
 - JavaScript
 - C#
-

5. Testing (පරීක්ෂා කිරීම)

Errors/Bugs identify කිරීම.

Types of Testing

Unit Testing

Single module testing.

Integration Testing

Modules together testing.

System Testing

Entire system testing.

User Acceptance Testing (UAT)

Users විසින් testing කිරීම.

6. Implementation (ක්‍රියාත්මක කිරීම)

System එක practical use එකට දීම.

Implementation Methods

Direct Changeover

Old system immediately replace කිරීම.

Parallel Changeover

Old and new systems together run කිරීම.

Pilot Changeover

Small section එකක first test කිරීම.

Phased Changeover

Step-by-step implementation.

7. Maintenance (නඩත්තු කිරීම)

System updates සහ bug fixing.

Maintenance Types

Corrective Maintenance

Errors fix කිරීම.

Adaptive Maintenance

Environment changes වලට adapt කිරීම.

Perfective Maintenance

Performance improve කිරීම.

Preventive Maintenance

Future problems prevent කිරීම.

SDLC Models

Waterfall Model

Linear sequential model.

Advantages

- Easy to understand
- Good documentation

Disadvantages

- Difficult to change requirements
-

Agile Model

Flexible development model.

Features

- Fast development
 - Continuous feedback
 - Team collaboration
-

Spiral Model

Risk-based development model.

Used For

- Large Projects
 - High Risk Systems
-

5. Database Deep Notes

දත්ත සමුදා

Database යනු කුමක්ද?

Organized collection of data.

Characteristics of Database

- Organized
 - Secure
 - Shared Access
 - Reduced Redundancy
-

DBMS (Database Management System)

Database manage කරන software.

Examples

- MySQL
 - Oracle
 - SQL Server
 - MongoDB
-

Advantages of DBMS

- Data Consistency
 - Multi-user Access
 - Backup & Recovery
 - Data Integrity
 - Security
-

Database Terms

English	Sinhala
Table	වගුව
Field	ක්ෂේත්‍රය
Record	චාක්රාව
Primary Key	ප්‍රාථමික යතුර
Foreign Key	විදේශීය යතුර

Primary Key

Unique Identifier.

Features

- Unique
- Cannot be null

Foreign Key

Tables connect කිරීම.

Normalization (සාමාන්‍යකරණය)

Data redundancy reduce කිරීම.

Benefits

- Save storage
- Reduce duplication
- Improve consistency

SQL Basics

SQL (Structured Query Language)

Database manage කිරීම සඳහා භාවිත කරයි.

SQL Commands

Command	Purpose
SELECT	Retrieve Data

Command	Purpose
INSERT	Add Data
UPDATE	Modify Data
DELETE	Remove Data

6. Networking Deep Notes

පරිගණක ජාල

Network යනු කුමක්ද?

Computers connect කර data share කිරීම.

Network Types

LAN (Local Area Network)

Small area network.

Example

- School Lab
-

MAN (Metropolitan Area Network)

City-wide network.

WAN (Wide Area Network)

Large geographical area.

Example

- Internet

LAN vs WAN

LAN	WAN
Small Area	Large Area
Fast	Slower
Cheap	Expensive

Network Devices

Router

Networks connect කරයි.

Switch

Devices intelligently connect කරයි.

Hub

Data broadcast කරයි.

Modem

Internet access ලබාදෙයි.

Access Point

Wi-Fi provide කරයි.

Switch vs Hub

Switch	Hub
Smart Device	Not Smart
Specific Data Send	Broadcast All Data
Fast	Slower

Network Topologies

Star Topology

Central device connection.

Advantages

- Easy troubleshooting
- Easy management

Disadvantages

- Central device failure affects network
-

Bus Topology

Single cable network.

Ring Topology

Circular network structure.

Network Protocols

TCP/IP

Main internet communication protocol.

HTTP / HTTPS

Web communication protocols.

HTTPS

Secure communication.

FTP

File transfer protocol.

SMTP

Email sending protocol.

IP Address Types

IPv4

Example:
192.168.1.1

IPv6

New advanced addressing system.

7. Cyber Security Deep Notes

සයිබර් ආරක්ෂාව

Types of Attacks

Virus

Self-replicating harmful program.

Worm

Automatically spreads through networks.

Trojan Horse

Useful software ලෙස පෙනෙන harmful program.

Phishing

Fake messages මගින් information steal කිරීම.

Ransomware

Files lock කර money demand කිරීම.

Security Methods

Firewall

Network traffic filter කරයි.

Encryption (සංකේතනය)

Data secure format එකකට convert කරයි.

Authentication

User identity verify කිරීම.

Password Security

Good Password Features

- Long Passwords
- Uppercase Letters
- Lowercase Letters
- Numbers
- Symbols

8. Web Technology Deep Notes

වෙබ් තාක්ෂණය

Website Architecture

Frontend

Visible part of website.

Technologies

- HTML
 - CSS
 - JavaScript
-

Backend

Logic and database handling.

Technologies

- PHP
 - Python
 - Node.js
-

APIs (Application Programming Interfaces)

Systems communicate කිරීමට allow කරයි.

Example

- Payment Gateways

Static vs Dynamic Websites

Static	Dynamic
Fixed Content	Changing Content
Simple	Interactive

9. Cloud Computing

වලාකුළු පරිගණනය

Internet-based services භාවිත කිරීම.

Types of Cloud Computing

SaaS (Software as a Service)

Example

- Google Docs
-

PaaS (Platform as a Service)

Developers සඳහා platform.

IaaS (Infrastructure as a Service)

Virtual infrastructure services.

Advantages of Cloud Computing

- Scalability
 - Remote Access
 - Cost Reduction
 - Backup Facilities
-

10. E-Government Deep Notes

විද්‍යුත් රාජ්‍ය සේවා

Government services online ලබාදීම.

Examples in Sri Lanka

- Online Tax Systems
 - Passport Applications
 - Vehicle Revenue Licenses
-

Advantages

- Transparency
 - Faster Services
 - Reduced Paperwork
 - Better Public Access
-

Frequently Asked Theory Questions

1. Explain SDLC with examples.
 2. Difference between RAM and ROM.
 3. Explain network topologies.
 4. What is normalization?
 5. Explain database relationships.
 6. Compare LAN, MAN, WAN.
 7. Explain functions of Operating System.
 8. Difference between Virus, Worm, Trojan.
 9. Explain Waterfall vs Agile Model.
 10. Explain frontend and backend technologies.
-

Final Exam Tips

- Definitions word-by-word පාඩම් කරන්න.
- Structured answers practice කරන්න.
- Flowcharts සහ diagrams practice කරන්න.
- Comparisons tables මතක තබාගන්න.
- Abbreviations revise කරන්න.
- Daily past papers practice කරන්න.
- SDLC, DBMS, Networks, Security වලට වැඩිපුර focus කරන්න.